Finite-population sample-size adjustment

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Population size N is finite.
- Initial large-population requirement n0 has been calculated.
- Need the final sample size for sampling without replacement.

2 DECISION / METHOD

1. Compute n0 using the ordinary mean/proportion formula.
2. Apply the finite-N adjustment.
3. ALWAYS ROUND UP the adjusted result.
4. Keep n0 and final n distinct.

3 FORMULA

```
n = n0*N / (n0 + (N-1))
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: this finite-population sample-size adjustment.
Still know: first calculate n0 correctly and round the adjusted n up.

5 EXCEL ROUTE

```
=ROUNDUP((n0*N)/(n0+(N-1)),0)
```

6 TINY WORKED EXAMPLE

If n0=385 and N=2000, adjusted n=322.99, so required n=323.

7 TOP EXAM TRAPS

- Applying FPC before n0 exists.
- Using current sample n where formula needs n0.
- Rounding down.
- Applying both SE-FPC and n-adjustment to the same planning step.

8 SAY ALOUD

First find the large-population n0, then shrink it for finite N and round up.

9 EXIT CHECK

Which value enters the adjustment numerator: current n or unadjusted n0?

ONE-GLANCE VISUAL ROUTE

